

Miss Around and Find Out: the Impact of Missing Data Types

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ABSTRACT

This report explores the impact of missing data on model performance using the **Heart Failure Prediction Dataset** [1]. We simulate three types of missing data (MCAR, MAR, and MNAR) and evaluate their effects. Various imputation techniques are applied, and their performance is compared to highlight the importance of handling missing data effectively in predictive modeling.

1 INTRODUCTION

Missing data is a data quality issue that can affect model performance. That is one of the reasons why we do data preprocessing: to detect and eliminate/mitigate these data quality issues. In this project, we will study three different types of missing data types: MCAR (Missing Completely At Random), MAR (Missing At Random), and MNAR (Missing Not At Random). After checking their effect on classification metrics against a control group, we will proceed with data imputation and check the performance of all 3 imputation techniques chosen (Mean, k-NN and MICE)[3].

2 METHODOLOGY

2.1 Dataset chosen

For this study, we use the **Heart Failure Prediction Dataset** [1], which contains various health parameters for patients diagnosed with heart failure. This dataset includes information such as age, sex, cholesterol levels, and heart rate, among other medical indicators. It is widely used for predictive modeling in healthcare care, particularly to understand factors related to heart failure outcomes. We selected this dataset since it didn't present any missing data in it, as well as there being a good balance in the classe features of the dataset.

2.2 Missing Data Types

- **MCAR(Missing Completely At Random)**: Missingness is independent of both observed and unobserved data.

$$P(M_i = 1|X_i, Y_i) = P(M_i = 1)$$

- **MAR(Missing At Random)**: Missingness depends on observed data but not unobserved data.

$$P(M_i = 1|X_i, Y_i) = P(M_i = 1|X_i)$$

- **MNAR(Missing Not At Random)**: Missingness depends on unobserved data (column itself).

$$P(M_i = 1|X_i, Y_i) \neq P(M_i = 1|X_i)$$

2.3 Imputation Techniques

- **Mean Imputation**: Missing values are replaced with the mean of the observed values for that feature.

- **k-NN Imputation**: Missing values are filled based on the values of the k -nearest neighbors. Proximity is calculated through the other features.
- **MICE Imputation**: Multiple datasets are created by iterating through features and imputing values using regression models. MICE captures the uncertainty of missing data by producing multiple imputed values for each missing entry.

3 RESULTS AND DISCUSSION

3.1 Control Variables

To simulate the three types of missing data, we create three distinct datasets using the `mdatagen` package [2]:

- **MCAR (Missing Completely At Random)**: We randomly remove 30% of the data from the column `ST_Slope` (x_{miss}), with no relation to other variables.
- **MAR (Missing At Random)**: Missing values are introduced in the `ST_Slope` column (x_{miss}) based on a condition in the 'Age' column (x_{obs}).
- **MNAR (Missing Not At Random)**: Missing data is removed from the column most correlated with the target variable.

Each dataset will have a missing rate of 30%, and all processes will be conducted with a fixed random seed of 42 to ensure reproducibility.

3.2 MCAR

These were the results obtained for the MCAR dataset:

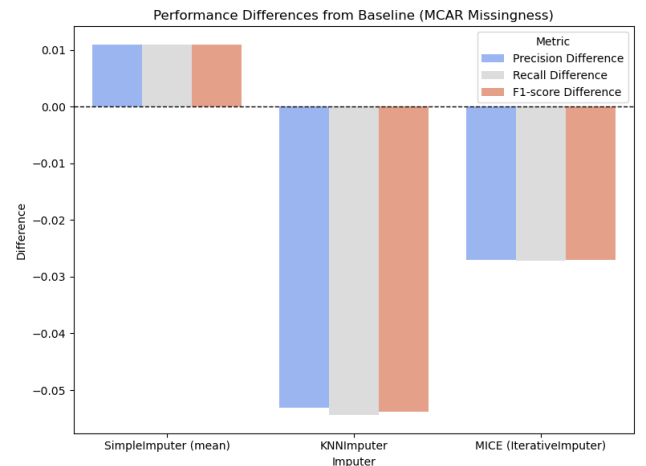


Figure 1: Comparison Imputers on MCAR Dataset

The **Simple Imputer** with the Mean strategy performed the best. This is intuitive, as the missing data is randomly distributed across the column, making the mean a reasonable replacement for the missing values.

The **k-NN** and **MICE** imputers performed much worse than the Mean strategy. This can be attributed to the fact that these imputers rely on patterns in the data, but with MCAR, the missingness is random and lacks any meaningful structure for the algorithms to capture.

3.3 MAR

These were the results obtained for the MAR dataset:

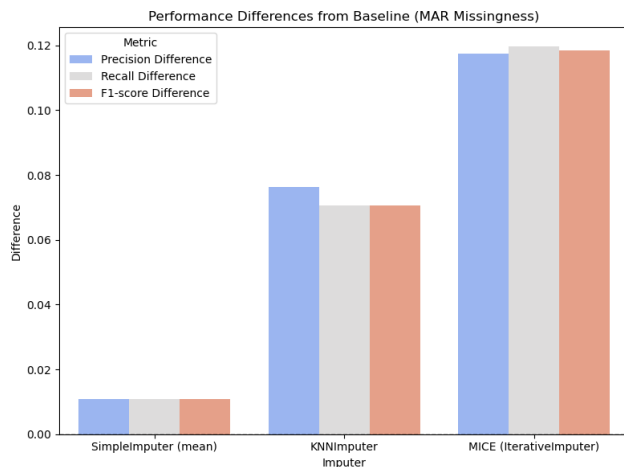


Figure 2: Comparison Imputers on MAR Dataset

The **Simple Imputer** with the Mean strategy had the worst performance. Since the missing data is related to the observed data, replacing missing values with the mean disregards the relationship between the missing and observed data, leading to suboptimal results.

Both **k-NN** and **MICE** imputers performed better than the Mean strategy. This is likely because these methods can utilize the observed data to create more informed imputations, capturing patterns that are relevant to the missing data.

The **MICE** imputer performed the best overall. Its iterative process was able to capture the relationship between the missing data and observed data (specifically, the Age column), resulting in the most accurate imputations.

3.4 MNAR

These were the results obtained for the MNAR dataset:

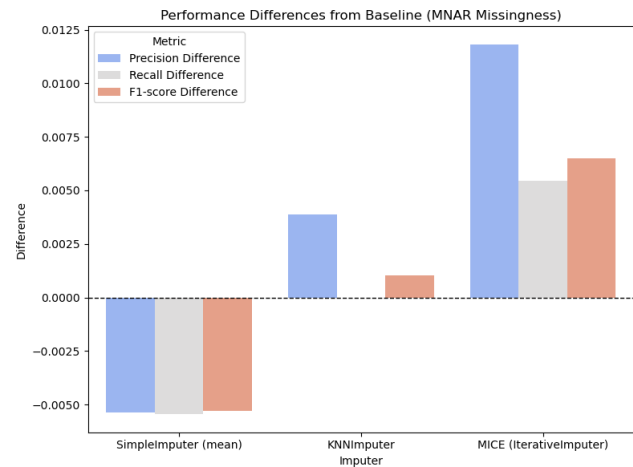


Figure 3: Comparison Imputers on MNAR Dataset

The **Simple Imputer** with the Mean strategy performed the worst. Since the missing data is related to the missing data itself, using the mean fails to capture any patterns, leading to poor performance.

Both **k-NN** and **MICE** imputers performed better than the Mean strategy. These methods are able to detect patterns in the missing data, leading to more accurate imputations.

However, the discrepancy in performance between the imputers was not as significant as in the other datasets. This suggests that for MNAR, other imputation methods or fine-tuning of hyperparameters could potentially improve results further.

4 CONCLUSIONS

In this study, we explored how the three types of missing data (MCAR, MAR, and MNAR) affect model performance and the importance of choosing the right imputation technique.

The results show that imputation methods significantly impact model accuracy and should be chosen based on the missing data mechanism:

For MCAR datasets, the **Simple Imputer** with the Mean strategy performed best, aligning with the randomness of the missing data. For MAR datasets, the **MICE** imputer outperformed others by capturing relationships between observed and missing values. For MNAR datasets, both **k-NN** and **MICE** performed well, though the performance difference was less significant, indicating room for hyperparameter tuning or exploring other methods. Overall, the study emphasizes the importance of understanding missing data mechanisms and selecting appropriate imputation techniques for optimal model performance.

REFERENCES

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